

Derivatives: Definition, Rules, Elementary Functions, and Inverse Trigonometric Derivatives

N-20221211

§1.1 Average velocity and instantaneous velocity

The first page starts from a graph and the idea of velocity. The average rate of change of f on $[x_0, x_0 + \Delta x]$ is

$$\frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}.$$

As $\Delta x \rightarrow 0$, the secant line approaches the tangent line if the limit exists. The instantaneous rate of change is the derivative:

$$f'(x_0) = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}.$$

Geometrically, this is the slope of the tangent line.

§1.2 Derivative notation

The note lists the notations

$$f'(x), \quad y', \quad \frac{dy}{dx}, \quad \frac{df}{dx}.$$

It also distinguishes the derivative at a point from the derivative function. The derivative at x_0 is a number; the derivative function assigns such a number to each point where the derivative exists.

The tangent line formula is

$$y - f(x_0) = f'(x_0)(x - x_0).$$

§1.3 Power functions

For positive integer n ,

$$(x^n)' = nx^{n-1}.$$

This follows from

$$(x + h)^n - x^n = \sum_{k=1}^n \binom{n}{k} x^{n-k} h^k,$$

and after dividing by h and taking $h \rightarrow 0$, only the $k = 1$ term remains.

The note extends the formula to rational powers by implicit differentiation. If

$$y = x^{m/n},$$

then

$$y^n = x^m.$$

Differentiating gives

$$ny^{n-1}y' = mx^{m-1},$$

so

$$y' = \frac{m}{n}x^{m/n-1}.$$

§1.4 Exponential and logarithmic derivatives

The page records

$$(a^x)' = a^x \ln a, \quad (e^x)' = e^x.$$

For logarithms,

$$(\ln x)' = \frac{1}{x}, \quad (\log_a x)' = \frac{1}{x \ln a}.$$

A useful derivation uses inverse functions: since $y = \ln x$ means $x = e^y$, differentiating gives

$$1 = e^y y' = xy',$$

so $y' = 1/x$.

§1.5 Trigonometric derivatives

Using the standard limits and addition formulas,

$$(\sin x)' = \cos x, \quad (\cos x)' = -\sin x.$$

Then quotient rule gives

$$(\tan x)' = \sec^2 x, \quad (\cot x)' = -\csc^2 x.$$

The reciprocal functions satisfy

$$(\sec x)' = \sec x \tan x, \quad (\csc x)' = -\csc x \cot x.$$

§1.6 Inverse trigonometric derivatives

For $y = \arcsin x$,

$$x = \sin y$$

and so

$$1 = \cos y y'.$$

Since $\cos y = \sqrt{1-x^2}$ on the principal branch,

$$(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}.$$

Similarly,

$$(\arccos x)' = -\frac{1}{\sqrt{1-x^2}}, \quad (\arctan x)' = \frac{1}{1+x^2}.$$

The note also derives inverse secant and inverse cosecant derivatives by drawing a right triangle and expressing the missing side length in terms of x .

§1.7 Examples

A typical example differentiates

$$y = \sqrt{x^2 + 1}.$$

By chain rule,

$$y' = \frac{x}{\sqrt{x^2 + 1}}.$$

Another example differentiates a quotient of trigonometric functions. The handwritten solution usually first rewrites the expression using identities, then differentiates. This is often simpler than applying quotient rule blindly.

§1.8 Context and outlook

This note builds the derivative table from a few principles: definition by difference quotient, binomial expansion for powers, inverse-function differentiation for logarithms and inverse trigonometric functions, and quotient/chain rules for combinations. The goal is not memorizing isolated formulas but recognizing which rule creates each formula.

§1.9 Derivative as first-order linearization

The derivative at a is best read as a first-order approximation:

$$f(a + h) = f(a) + f'(a)h + o(h).$$

The number $f'(a)$ is the linear map that best approximates the change of f near a . In one dimension, linear maps are multiplication by a number, so this viewpoint collapses to the familiar tangent slope.

This explains why the chain rule has the form it does. If

$$f(a + h) = f(a) + f'(a)h + o(h), \quad g(b + k) = g(b) + g'(b)k + o(k), \quad b = f(a),$$

then

$$g(f(a + h)) = g(f(a)) + g'(f(a))f'(a)h + o(h).$$

Thus

$$(g \circ f)'(a) = g'(f(a))f'(a).$$

The chain rule is composition of linear approximations.

§1.10 Inverse trigonometric derivatives and the inverse function theorem

The inverse-function formula

$$(f^{-1})'(y) = \frac{1}{f'(x)}, \quad y = f(x),$$

is a one-dimensional form of the inverse function theorem. If f is continuously differentiable near x_0 and $f'(x_0) \neq 0$, then f has a local differentiable inverse near $f(x_0)$.

For $y = \arcsin x$, the equation $x = \sin y$ is inverted only after restricting sine to

$$[-\pi/2, \pi/2].$$

On this branch, $\cos y \geq 0$, so

$$1 = \cos y \frac{dy}{dx}, \quad \frac{dy}{dx} = \frac{1}{\cos y} = \frac{1}{\sqrt{1-x^2}}.$$

The square root sign is not arbitrary; it comes from the chosen branch of the inverse.

§1.11 Endpoint blow-up and failure of local invertibility

The derivative

$$(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$$

blows up as $x \rightarrow \pm 1$. Geometrically, near $y = \pm\pi/2$, the graph of $\sin y$ has horizontal tangent:

$$\cos y = 0.$$

The inverse function theorem fails exactly where $f'(y) = 0$. The inverse may still exist globally on a restricted interval, but it is no longer differentiable with a finite derivative at the endpoint.

This is the conceptual reason inverse-trigonometric formulas have domain restrictions and singular denominators.

§1.12 Logarithm as inverse and as group homomorphism

The derivative of $\ln x$ follows from $x = e^y$:

$$1 = e^y y' = xy', \quad y' = \frac{1}{x}.$$

But $\ln$ also has a structural meaning:

$$\ln(xy) = \ln x + \ln y.$$

It converts multiplication in $(0, \infty)$ into addition in $\mathbb{R}$. Its derivative $1/x$ is the infinitesimal version of multiplicative scaling.

In Lie-group language, $(0, \infty)$ is a one-dimensional Lie group under multiplication, and $\ln$ identifies it with the additive group $\mathbb{R}$. The elementary derivative formula is the local shadow of this group isomorphism.

§1.13 Automatic differentiation and computation graphs

The product, quotient, and chain rules are the mathematical basis of automatic differentiation. A complicated expression is a computation graph. Each node knows its local derivative, and the chain rule propagates derivative information through the graph.

For example, if

$$y = \sqrt{x^2 + 1}, \quad u = x^2 + 1, \quad y = \sqrt{u},$$

then

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx} = \frac{1}{2\sqrt{u}} \cdot 2x = \frac{x}{\sqrt{x^2 + 1}}.$$

The ordinary chain-rule computation is already reverse or forward mode automatic differentiation in miniature.

§1.14 Jets and higher-order local data

The k -jet of a function at a records its derivatives up to order k :

$$j_a^k f = (f(a), f'(a), \dots, f^{(k)}(a)).$$

The Taylor polynomial is the coordinate expression of this finite local data:

$$T_k f(a + h) = \sum_{j=0}^k \frac{f^{(j)}(a)}{j!} h^j.$$

The first derivative is therefore not an isolated operation; it is the first nontrivial layer of a tower of local approximations.

§1.15 Differentials and one-forms

The notation dy/dx is sometimes taught as a fraction-like symbol. A safer advanced interpretation is through differentials. For $y = f(x)$,

$$dy = f'(x) dx.$$

Here dx is the basic one-form on the line, and dy is its pullback through f . The chain rule becomes the rule for pulling back one-forms:

$$(g \circ f)^*(dz) = f^*(g^*(dz)).$$

This gives a structural meaning to the notation without treating infinitesimals as informal numbers.

§1.16 Derivatives as linearization and pullback

The derivative table is built from a few structural principles: linear approximation, inverse functions, branch choices, composition, and local Taylor data. The formulas for powers, logarithms, trigonometric functions, and inverse trigonometric functions are not separate facts; they are examples of how local linear behavior is transported through algebraic operations and inverses.